Empirical Runtime Analysis

COP 3502C – Computer Science I

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Recall

- Count how many values are common between two sorted lists

- We explored 3 strategies that have different running times

Scenario

$N = 1$

- You can run an experiment and collect data for analysis
- Given the input size, record the time elapsed

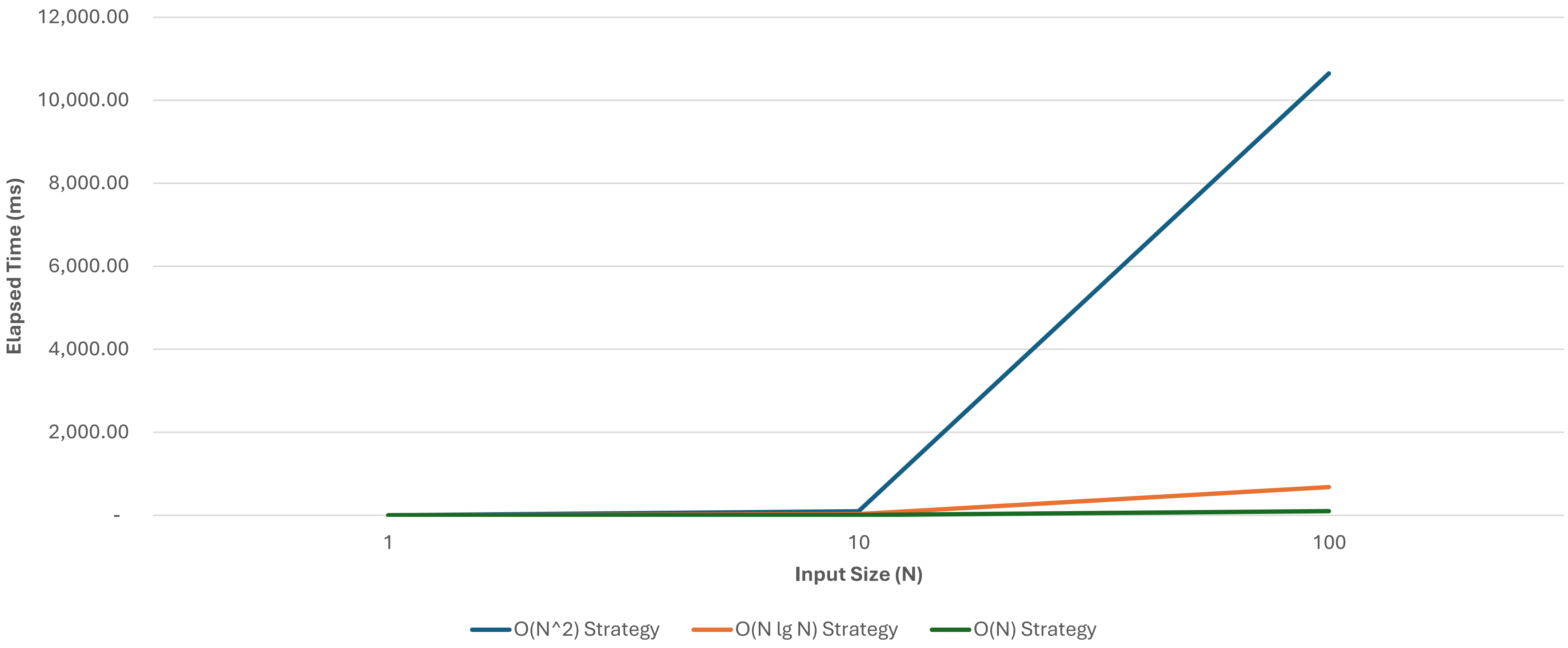
$N = 1000000$

- Note that this should be an average of at least 5 runs per input size

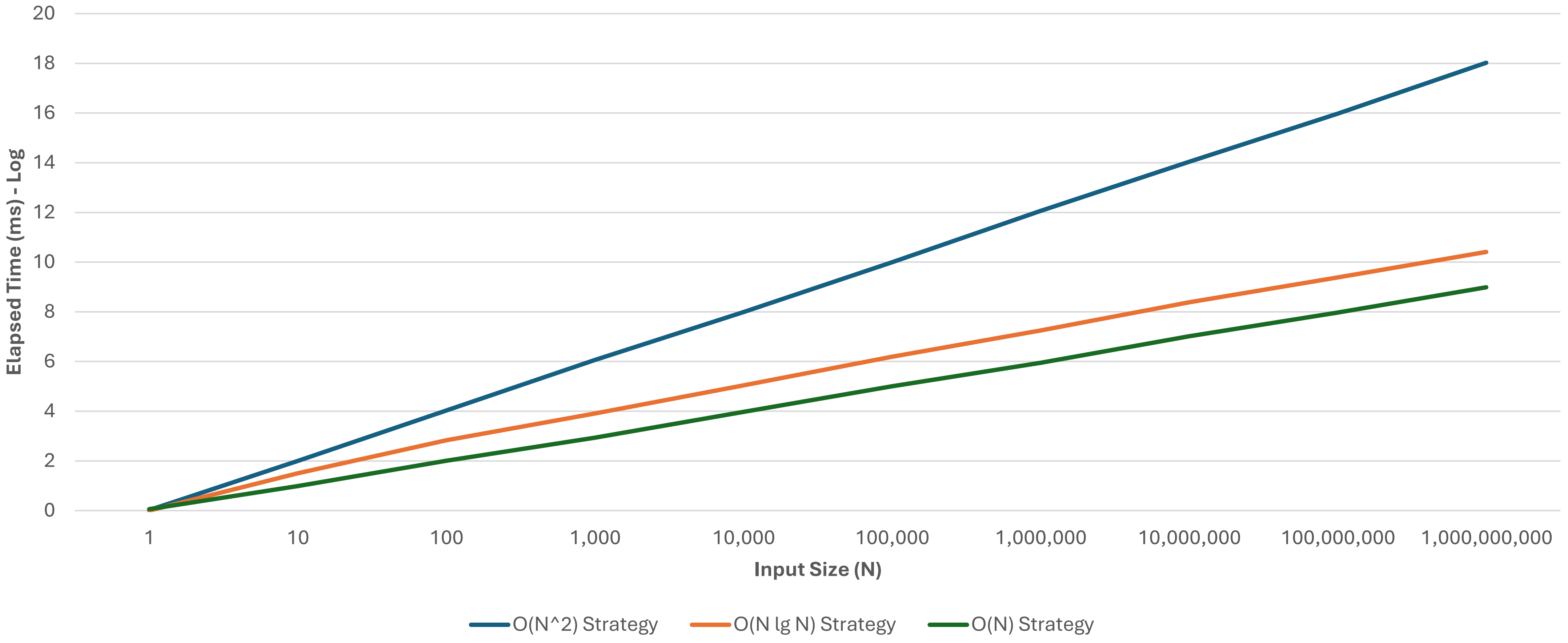
Experimental Results

N	O(N^2) Strategy	O(N lg N) Strategy	O(N) Strategy
1	1.05	0.00	1.15
10	98.62	31.67	9.77
100	10,647.69	680.46	100.68
1,000	1,152,302.99	8,059.05	857.53
10,000	97,658,466.25	109,956.91	9,455.62
100,000	9,765,863,043.05	1,567,570.11	101,109.23
1,000,000	1,157,921,281,550.73	17,912,837.28	884,900.64
10,000,000	107,674,347,291,529.00	239,842,315.94	10,375,698.02
100,000,000	9,530,525,614,065,040.00	2,416,231,220.93	93,993,613.10
1,000,000,000	1,054,256,004,358,590,000.00	25,674,938,644.41	970,830,625.02

Visualization



Visualization (Log)



$T(n) = O(n) \Rightarrow T(n) = c \cdot n$

O(N) Strategy: Empirical vs. Theoretical /1

N	Empirical $T(n)$	Theoretical $O(n)$	C_N
1	1.15	1	**1.147**
10	9.77	10	0.977
100	100.68	100	1.007
1,000	857.53	1,000	0.858
10,000	9,455.62	10,000	0.946
100,000	101,109.23	100,000	1.011
1,000,000	884,900.64	1,000,000	0.885
10,000,000	10,375,698.02	10,000,000	1.038
100,000,000	93,993,613.10	100,000,000	0.940
1,000,000,000	970,830,625.02	1,000,000,000	0.971

O(N) Strategy: Empirical vs. Theoretical /2

N	Empirical	Theoretical	Recomputed
1	1.15	1	1.15
10	9.77	10	11.47
100	100.68	100	114.70
1,000	857.53	1,000	1,147.00
10,000	9,455.62	10,000	11,470.00
100,000	101,109.23	100,000	114,700.00
1,000,000	884,900.64	1,000,000	1,147,000.00
10,000,000	10,375,698.02	10,000,000	11,470,000.00
100,000,000	93,993,613.10	100,000,000	114,700,000.00
1,000,000,000	970,830,625.02	1,000,000,000	1,147,000,000.00

Notes

- Identify the constant C_N for each of your N values
- C_N is simply computed as empirical divided by theoretical
- Find the max C_N value found because we want the **upper bound**
- Multiply that constant with the other theoretical times

Discussion

$T(n) = O(f(n))$

$T(n) = c \cdot f(n)$

$T(n) \to$ empirical time

$c \to$ max constant found

$f(n) \to$ theoretical runtime

Practice Problem /1

Suppose we have an $O(n^3)$ function that took 3 seconds to execute with input size n=5,000. What would you expect the runtime be if n=10,000?

$$T(n) = \underline{O}(n^3)$$

$$T(n) = c \cdot n^3$$

$$\frac{3}{5000^3} = \frac{c\,(\cancel{5000^3})}{\cancel{5000^3}}$$

$$\boxed{\frac{3}{5000^3} = c}$$

$$T(n) \overset{?}{=} \underline{c} \cdot (10000^3)$$

$$= \frac{3}{\cancel{5000}^3} \left[\overset{2^3}{\cancel{10000}^3} \right]$$

$$= 3 \cdot 2^3$$

$$= 3(8)$$

$$\boxed{T(n) = 24}$$

Practice Problem /2

An algorithm processing a two-dimensional array with R rows and C columns runs in $O(RC^2)$ time. For an array with 100 rows and 200 columns, the algorithm processes the array in 120 ms. How long would it be expected for the algorithm to take when processing an array with 200 rows and 500 columns?

$$T(n) = O(RC^2)$$

$$T(n) = c \cdot (RC^2)$$

$$\frac{120}{100 \cdot 200^2} = \frac{c \cdot (\cancel{100 \cdot 200^2})}{\cancel{100 \cdot 200^2}}$$

$$\boxed{\frac{120}{100 \cdot 200^2} = c}$$

$$T(n) = c \cdot (200 \cdot 500^2)$$

$$= \left[\frac{120}{\cancel{100} \cdot 200^2}\right]\left[\cancel{200}^{2} \cdot 500^2\right]$$

$$= \frac{120 \cdot 2 \cdot \cancel{500}^{2}\,5^2}{\cancel{200^2}\,2^2}$$

$$= \frac{120 \cdot \cancel{2} \cdot 5^2}{2\,\cancel{2}}$$

$$= \frac{120 \cdot 5^2}{2}$$

$$= 60 \cdot 5^2$$

$$= 60 \times 25$$

$$\boxed{T(n) = 1500 \text{ ms}}$$

30
60
25
300
120
1500

Practice Problem /3

Querying a user in our database of 10^4 users takes about 10 ms. The runtime of the query is logarithmic with respect to the number of users. Namely, if there are n users, a query takes $O(\log_2 n)$ time. About how many users can we support while taking no more than 20 ms per query.

$$T(n) = O(\lg n)$$

$$T(n) = c \lg n$$

$$\frac{10}{\lg 10^4} = \frac{c \cancel{\lg 10^4}}{\cancel{\lg 10^4}}$$

$$\boxed{\frac{10}{\lg 10^4} = c}$$

$$T(n) = c \lg n$$

$$20 = \left[\frac{10}{\lg 10^4}\right] \lg n$$

$$\frac{20 \lg 10^4}{10} = \lg n$$

$$2 \lg (10^4) = \lg n$$

$$\lg (10^4)^2 = \lg n$$

$$\lg 10^8 = \lg n$$

$$\boxed{10^8 = n}$$

$$(10^4)^2 = 10^8$$

Practice Problem /4

Suppose we have an $O(n^2)$ function that took 12.4 seconds to execute with n=1,000. We ran it again and the runtime was approximately 49.6 seconds. What was the approximate input size the second time around?

$$T(n) = O(n^2)$$

$$T(n) = c n^2$$

$$\frac{12.4}{1000^2} = \frac{c \cdot \cancel{1000^2}}{\cancel{1000^2}}$$

$$\boxed{\frac{12.4}{1000^2} = c}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

$$T(n) = c \cdot n^2$$

$$49.6 = c \cdot n^2$$

$$49.6 = \frac{12.4}{1000^2} \cdot n^2$$

$$\frac{49.6\,(1000^2)}{12.4} = n^2$$

$$\sqrt{4\,(1000^2)} = \sqrt{n^2}$$

$$\sqrt{4}\,\sqrt{1000^2} = n$$

$$2 \cdot 1000 = n$$

$$\boxed{2000 = n}$$

$$\begin{array}{r} 248 \\ +124 \\ \hline 372 \\ +124 \\ \hline 496 \end{array}$$

Final Notes

- Essentially there are two kinds of questions given that you know the **running time complexity** (i.e., **Big Oh**) of your program

- If my input size is **`n`**, how long will it take for the program to run?

- If I only have **`T`** amount of time, how big should my input size be?

- Use these to inform your decision-making if you have **constraints**

Questions?